

Data, Indicators and Decision Making in Afghanistan
Relief and Rehabilitation Programs

Shawn E. Messick

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1. SUMMARY

The purpose of the consultancy was to identify data, indicators and mechanisms for compiling and maintaining the requisite data flows to support the decision making process for program coordination, implementation and monitoring. ProMIS is the focal point for assembling and distributing such products and services. At its heart ProMIS is built on a georeferenced database structure and is therefore a geographic information system (GIS). Hence this project has focused the identification of data, formats and organizational mechanisms and networking that should support and promote data collection, sharing and indicator analysis.

1.1. CHICKEN OR THE EGG

Introducing new information management practices with the aim of improving the way organizations work is a bit like asking whether the chicken or the egg came first. If you don't know what a GIS can do differently, chances are you aren't going to ask it to do anything different. Very few products produced by ProMIS (using other agencies data) or other agencies with in-house GIS capabilities move beyond straight cartography into spatial analysis. In part this is a reflection of the ad hoc nature of current relief and reconstruction efforts and partly the lack of agency or sector standards or policies. Presented with such products, most managers would not be impressed enough to commit their staff to a lengthy data conversion process. However, such a reaction is misplaced and akin to shooting the messenger. The poor products are an accurate reflection of how agencies and NGOs think about what they are doing. In bringing about change the primary users must be educated and trained in the tools and opportunities for improving the effectiveness and efficiency of what they do. Of equal or greater interest to relief and rehabilitation efforts are the abilities inherent in GIS tools to generate very compelling depictions of problems, needs and justification for continued funding. There is a need for selling and educating potential users on the abilities of GIS because computers don't make decisions, people do. Correcting this problem requires the appreciation that a decision makers understanding is as important to a good decision as good data. Managers would rather live with a problem they can't solve than apply a solution they don't understand. Hence, focus on indicators is a fruitless effort if the intended users don't know how to use them or understand the tools for manipulating them. Findings that demonstrate such problems include:

- Most organizations maintain no direct connection between information collected and program delivery requirements or other sector standards.
- Sector specific standards for service delivery are generally lacking (most do not exist), nor is there a focus on basic elements of information.

To address these failings requires honest discussion among sector agencies regarding what is important to collect. In doing so they should keep in mind Occam's Razor. The principle is attributed to the medieval philosopher William of Occam. The principle states that one should not increase, beyond what is necessary, the number of entities required to explain anything. To reach this end it is recommended that:

- Sector agencies and RCOs should collaborate with their partners in compiling countrywide datasets on all infrastructure and service delivery/access/entitlement issues. These sector specific databases will provide the context by which needs, requirements, impact and progress can be measured.

- ProMIS should collaborate with Sectors and UNOCHA/SMU to conduct workshops aimed at identifying specific standards, essential data elements, the analytic means for evaluating the data and meaningful classes for displaying the results.

1.2. A BIRD IN HAND...

Information management in support of decision-making in the Afghan relief and rehabilitation programs suffers many of the same problems experienced in other similar programs. Of the observed findings, very few relate to technology issues. Rather they relate to the lack of effort and understanding of organizations and managers to take the necessary steps towards adopting basic information management standards and procedures. Looking to ProMIS to solve these problems is unrealistic. ProMIS does not, nor should it, have the sector technical expertise to tell agencies what is important or not. While this report contributes to the process of facilitating data collection and indicator analysis, the primary responsibility for information collection resides with the NGOs and Agencies comprising each program sector. Findings directly related to this issue include:

- Surveys and data compiled in the last 10 years have to a large extent been lost.
- Sector lead agencies (at HQ and regions) and RCO efforts at data collection are not coordinated, standardized or responsive to each other's needs.
- Countrywide information exists yet is not being compiled, instead new surveys are commissioned.
- After 12 years of programs there is still no single useable information source that details organization activity locations down to the village level.

The first step in addressing these findings is compiling what is known. Completing this modest step will set in motion all follow on analyses. Recommended actions include:

- ProMIS should work with RCOs, Sector agencies, ACBAR and NGOs in compiling a standard, settlement level, georeferenced organizational activity location (who's doing what where) database.

1.3. MANAGING CHANGE

A major constraint on the effective functioning of indicator-based analysis is the lack of upper management endorsement and promotion of information standards. Similarly, sector lead agencies have been unable to separate sector management responsibilities from internal project management interests. This has resulted in a primary focus on areas of interest to the agency and limited focus on defining a common operational picture for the sector. Furthermore, the lack of a country wide focus on collecting data is the primary reason that there is very little credible information to answer basic questions such as: where is the need, what resources are required, what has been the impact, and what progress has been made? The phrase "garbage in, garbage out" defines the current state of affairs. Specific findings in this theme include:

- Lack of UNCO or Donor (APB) support or influence promoting information management and reporting standards.
- Organizations focused on point-of-service implementation vs. service delivery efficiency.
- There is widespread bickering amongst agencies and NGOs on data quality.
- There is significant redundancy in information collection efforts.

- Sector lead agencies have serious limitations in their ability to handle information.

There are no overnight solutions to these problems. However, some immediate actions can be taken that will start things moving in the right direction. These include:

- An Information Standards Committee (UNOC/APB level), as originally proposed in the ProMIS project documentation, be constituted and convened regularly to review and promote information management requirements, address data entry bottlenecks, reduce redundancy and track sector efforts of standardizing information requirements, compilation of member data sets and setting policies and standards.
- As a matter of urgency, UNCO/APB should determine which data set will be used – the 29 or 32 province geocode sets. On this decision rests the entire structure of information compatibility and sharing.
- ProMIS should prioritize efforts on the maintenance and updating of base map layers (Province, district, village) and geocodes and facilitate dissemination to field users.

1.4. CONCLUSION

Managing change does not have to be difficult. The concept and practice of Principled Common Programming and Thematic groups adopted by the Afghan programs is a first step. Implementing a credible information management system can provide this process with the necessary support to realize the interconnections between programs and make the change a substantive reality for the participants, rather than a repackaging of old practices.

To make an indicator analysis and data collection system work, management must claim ownership, promote standards, and make it a central tenet of the management and coordination process. It has been said that trying to coordinate NGOs and UN agencies is akin to trying to herd cats. With the right inducement, leading can be more effective in promoting independent organizations to move in the right direction. Effective information management and indicator analysis can provide organizations with a common operational picture and the necessary inducement to engage in coordinated and cooperative efforts.

2. INTRODUCTION:

2.1. BACKGROUND

The overall objective of the ProMIS project is to enhance the capacity of UN Agencies, NGOs and donors to undertake analysis and planning for operational and emergency activities, and common programming for Afghan rehabilitation and development activities. To achieve this objective ProMIS maintains a number of systems and services to share data, documents, spatial layers, and other information for the aid community in Afghanistan.

The ProMIS tools consist of four components: an information management system, a metadata system (which provides information about the source, content and quality of data sets), a web information access system and a model core database. The prototype information system is based on geographic information system (GIS) technology that enables the presentation and analysis of spatial data. In other words, basic and sectoral data are organized geographically and are presented in the form of maps or tabular information based on geographical units.

Common programming in Afghanistan has now taken a thematic, as opposed to a sectoral, approach. The themes are defined in the UN Consolidated Interagency Appeal as: Alleviation of human suffering; human rights; the provision of social services; sustainable livelihoods; and, the return of refugees. Basic background data on Afghanistan, organized under these themes, is required to support both the design of programs and projects that support this approach as well as measure the impact of such programs. Background data include basic information on the location of schools, health clinics, number of trained professionals, and so on. Background data will also include the policy objectives, measures and standards by program, project, sector and theme. The ProMIS tools will facilitate access to this information.

The deployment of the ProMIS tools will require the organization of mechanisms to maintain timely data sets as well as ensure that data continues to support policy decision support objectives. Currently data is collected on an individual basis by agencies and NGOs with little to no compiling/consolidation at the sector or central levels. The consultant will be responsible for both identifying sources of baseline data and for producing standards and guidelines for managing the incorporation of this information into the ProMIS core database.

2.2. TERMS OF REFERENCE FOR THE CONSULTANT

Based in Islamabad the consultant will:

- Consult with individuals in UN Agencies and NGOs to develop a basic inventory of baseline data requirements to support the thematic-centered programming in Afghanistan;
- Identify mechanisms for collecting these data , incorporating what is currently collected by agencies and NGOs;
- Conduct a survey of existing background information on Afghanistan to provide a historical context.

2.3. APPROACH

The Consultancy was performed during the period 5 September to 20 October 2000. The consultant visited organizations in Islamabad and Peshawar in Pakistan and Surveyed, by email, UN Regional Coordinators in Afghanistan.

The consultant used a systems management approach requiring individual interviews with information management and Project/program management staff in various agencies. The consultant reviewed with agency data collectors the intended use of what they have collected. Specifically the consultant focused on how the collected data was going to be analyzed for programming purposes. Determining essential data and types of analysis is in keeping with efforts such as the Sphere Project and WHO's Data for Decision Making effort. Through this process the users' knowledge of analytic techniques and information management issues were also evaluated.

Such an approach necessarily focuses on the human and organizational aspects of information collection, sharing and analysis. Based on the consultant's experience in other relief and development programs, issues surrounding transaction costs were also highlighted as primary impediments for information sharing.

2.4. ACKNOWLEDGEMENTS

The consultant would like to thank the staff of the ProMIS office for their unwavering support and candid observations on problems and improvements in the functioning of the ProMIS project. Additional thanks go to the FAO Afghanistan Office for their timely and efficient administrative support during the mission. Lastly I would like to extend my thanks to all those individuals in UN Agencies, NGOs and the UN Coordinator's Office who made time to talk critically about what they do.

3. FINDINGS

3.1. CONDUCT A SURVEY OF EXISTING BACKGROUND INFORMATION

3.1.1. Findings:

- Surveys and data compiled in the last 10 years have to a large extent been lost.
- Countrywide information exists yet is not being compiled, instead new surveys are commissioned.
- After 12 years of programs there is still no single useable information source that details organization activity locations down to the village level.

3.1.2. Discussion

There is a dearth of information available on Afghanistan from the last ten years. In part this is a result of a failure to maintain copies of databases created. While paper bound reports exist, these provide only summary data. The ACBAR (Agency Coordinating Body for Afghan Relief) library in Peshawar does maintain a large repository of UN and NGO reports and studies as well as significant numbers of former Afghan governmental publications. ACBAR also publishes a three-volume agency activity database report on an annual basis. Minor modifications would greatly enhance its usability. The only historical data sets that have been in continuous use are the demographic data sets comprising census data from the 1970's and 1980's as well as those conducted by USAID and the UN UNIDATA project in the early 1990's. Agricultural land cover data in GIS format is available from FAO sponsored satellite surveys from 1972 and 1993.

Two very large projects conducted by USAID in the early 1990's exemplify the problem. USAID commissioned Development Alternatives International (DAI) to compile digital mapping of Afghanistan in 1994. Very little of the data is available except for catalogues of datasets that were compiled. I was able to locate and copy an incomplete set of data that had been converted from a Bernoulli drive retained by a former staff member¹. The information comprises 130MB of compressed data (with 2,198 separate files in 67 folders) in ArcPlot and ArcInfo coverages that will require conversion and reconstitution to evaluate the extent of usefulness. Similarly the University of Nebraska at Omaha (US) conducted an exhaustive survey on the *Status of Education in Afghanistan*. The survey inventoried all of the primary and secondary schools in Afghanistan (name, village, district, province), covering male and female enrollment and enrollment ratios, teachers, textbooks, facilities as well as progression and drop-out rates. While the survey was conducted in 1994, no copies of the database appear to have remained in country. No paper printouts of the data tables appear to have been retained. The only record of the study is a large summary report.

Most organizations UN and NGO have database information starting from 1996. Most of this data is project specific and inventory in nature. A number of baseline surveys have been commissioned by NGOs and the UN focusing on districts, summary reports are available but very little of the data is available at the village level in tabular

¹ The staff member at his own expense had sent the drive to a data conversion company in the US to have the contents extracted to a CD.

format. The data have been aggregated on survey issues thereby losing all geographic specificity.

One of the easiest and yet most elusive goals of relief operations is to capture who is doing what where. While ACBAR produces an annual report of NGO activities in Afghanistan, these are mostly in summary form and contain only the most general of geographic specificity. I have ascertained that most organizations have kept a record on computer or paper of their activities to date. Many Afghan field staff also have a large store of information in memory about what projects were done where, which needs to be documented. Two large long-term NGOS have compiled significant datasets Swedish Committee for Afghanistan (SCA) and Danish Committee for Aid for Afghan Refugees (DACAAR). DACAAR provided me an extract of their well data that covers some 19,000 wells in Afghanistan. SCA also have a fairly extensive database. These organizations are willing to contribute to a countrywide coverage of Afghan data providing that a logical structure is provided and guarantees on are in place on how the data is used. Such requirements are consistent with standard data sharing agreements.

3.1.3. Conclusions

Historical data is of marginal use given the requirement for extensive manual review of narrative documents to extract pertinent data of limited immediate value. However, large data sets do exist for specific sector areas of interest. Applying standard formats and common referents (geocodes) to such data sets help organizations to define what has or will be done. Inventorying such data is a necessary first step towards implementing indicator analysis. Issues regarding compilation will be addressed in section 4.3.

3.2. DEVELOP A BASIC INVENTORY OF BASELINE DATA REQUIREMENTS

3.2.1. Findings:

- Most organizations maintain no direct connection between information collected and program delivery requirements or other sector standards
- Sector specific standards for service delivery are generally lacking (most do not exist), nor is there a focus on basic elements of information.
- Sector lead agencies (at HQ and regions) and RCO efforts at data collection are not coordinated, standardized or responsive to each other's needs.
- Organizations focused on point-of-service implementation vs. service delivery efficiency.

3.2.2. Discussion

Many of the reports reviewed are simple recapitulations of collected data. Aside from percentages, little to no "analysis" has been performed on the data. Similarly, representations of the data in charts and thematic map products provide class breakdowns that are at worst arbitrary and at best simple equal range breakdowns (e.g. 1-10, 11-20, 21-30 etc). Such class breakdowns are very misleading. The viewer's eye is drawn to the darker hues, interpreting a suspected value in an implied

scale of magnitude that does not exist in the underlying data. Such is the strength and weakness of geographic information systems (GIS) products. Poor representation of data can lead the viewer to interpret a message that was never intended. If assembled correctly the viewer of graphic data (charts, thematic maps) should come away with an understanding of the issues and the messages that the producing organization intended. Only a few of the organizations in Afghanistan have made this conceptual leap. Products generated by ProMIS based on agency provided data share this shortcoming. This is not a fault of ProMIS, it does not have sector technical expertise, hence it relies on what the contributing agency provides.

In reviewing earlier ProMIS consultations with agencies regarding the issue of baseline data and indicators I came across multi-page documents concerning everything from hides produced/sold to neonatal deaths. Such wish lists burden data collection efforts with requirements that have little immediate use. Similarly, while reviewing a number of agency and regional coordination databases, existing and proposed, as well as survey formats, the overwhelming question raised was WHY? Why were all of these information elements being collected? What did the collectors intend to do with them? Meetings with Program/Project Managers and Information Specialists raised similar questions. Questions of “how do you intend to use this data” resulted in a stock answer of “to better inform our programming”. When re-phrased as “how will you analyze this data, what meaning do you seek to find by collecting it and what is the direct connection between this piece of information and a project goal or resource allocation requirement?” answers were very unclear. The only efforts that have a definitive link between information collection and project decision-making are the MAPA mine action database, the WFP VAM Emergency Food Needs Assessment and the WHO/Health Information System Task Force Effort (Guidelines published in September 2000).

Why do we collect information and how do we intend to use it for decision-making is at the crux of defining baseline information and indicators. With the exception of a very few cases, indicators are derived values. By derived I mean that some action has been performed on the raw data to render it comparable to values for similar units of analysis. For example, some of the simplest analyses normalize data over population to yield per capita values. While some per capita values are useful as indicators, other analyses require a second process be applied to compare these values against a desired value.

Other types of indicators are signal events that have been determined based on analysis of a situation. For example, the arrival of internally displaced persons (IDP) at a location may signal a problem in the area of origin and a requirement to have sufficient resources to meet this new population's needs. For the drought in Afghanistan, evaluating the extent or failure of winter wheat planting may give an advance indication of potential new IDPs in the spring and summer. However, while the completion of a revised curriculum may be a step towards project completion of an education assistance program, a better measure of success is the degree to which the physical textbooks have been delivered into the hands of the student as well as the degree to which teachers have been trained in delivering the new curriculum. To be effective, indicators must be linked singly or in combination to decision requirements.

If an indicator must be a value or event that has been derived through analysis, how can we characterize the other information that forms the bulk of most surveys? I call these elements “nice-to-know” information. Such information can provide color, background and other contextual information. These information elements, if well thought out, can also form the basis for more advanced analyses. However, given limited information collection resources, and when trying to build a cooperative network of information providers based on organizations that have a strong independence of action ethic, extraneous information requirements are not politic.

To derive an indicator from information requires that an action be taken on the element of information. Hence, where do we find such operators? The most common locations are policies and standards. These policies and standards must be specific.

Standards should:

- Identify measurable long-term goals or desired end-state
- Facilitate the identification of short-term objectives in the context of long term goals
- Allows comparative analysis between units (community, District, Province)
- Facilitate measures of progress
- Facilitate forecasting and relief/development transition links

There are two common denominators to most analyses. Generally, relief programs are entitlement based. That is they are service delivery to a population, or portion thereof, in a specific location. When asking questions such as what are the needs and how well have they been addressed, demographic and geographic location data are essential.

Demographic data refers not only to headcounts but also to profiles. However, some cautions need to be exercised in compiling profiles. A common use of profiles are to extrapolate population figures based on a discrete subset such as children under five years of age. For example, a standard population profile states that the under five population generally constitutes 20 percent of a population. Hence an observed count of 20 children under five in a community would yield a population of 100 for the community. War affected societies have generally sustained population losses due to combat or depredations of civilian population. Hence it is important for agencies working in areas to assess a demographic profile for that area. One demographic element frequently used in relief planning is the female headed house hold. This is often only reported as a percentage of all households. In this case the head count is required. It is not likely that population growth will generate a proportionate increase in female headed households. See annex B for a discussion on demographic data issues.

Geographic data refers to units of analysis. For Afghanistan this is limited to province, district and settlement. I referred to settlement as opposed to village because villages in Afghanistan can also comprise small outlying settlements. Existing data sets and gazetteers do not provide such a breakdown of association, hence there is no means of currently displaying such community relationships until the required information is collected and reported from the field. Using the 1979 Locality Gazetteer for Afghanistan (LGA) published by the Afghan Central Statistics Office (CSO) each province, district and settlement was allocated a unique code. Generally geocodes cascade. For example Kabul province is 01, Kabul district is 0101 and

Kabul City is 0101001. This means that Kabul city is the first settlement in Kabul district, which is the first district in Kabul province, which is the first province in Afghanistan. Such geocodes are a well established practice internationally and allow the creation of a unique record for a defined unit in geographic space. If we know the location then a name or multiple names for that unit can be assigned. The geocodes help overcome confusion resulting from different spellings and different place names that normally stymie efforts at compiling data sets from multiple sources. In this sense the geocodes become the common referent for data collection.

While ProMIS has undertaken to promulgate these geocodes, many organizations do not perceive any benefit to them for undertaking a large manual coding process. Such perceptions directly relate to the lack of analytic standards and objectives and knowledge of the benefits of such analyses. However, where organizations have embraced the geocodes, many have commented on discrepancies such as non-existent settlements, new settlement names or many new or old settlements that do not have geocodes. By prior local management decision, ProMIS has not adjusted the base map to reflect these new features. ProMIS has no feedback form for standardizing how corrections to the base map can be submitted. This failure on the part of ProMIS has created frustration and ill will toward ProMIS and has damaged the credibility of the ProMIS information management effort.

The most well developed sets of indicators come from the epidemiological community. For most diseases and health problems, standard practice is to calculate rates for morbidity and mortality. These rates are calculated by taking the number of reported cases, prorated over the duration of collection, deriving a per capita value and then multiplying by a factor of 10,000 or 100,000. The second step in the process then compares this value against a standard. For example many diseases have country specific historical rates that represent the baseline. By comparing the derived value to the standard we are able to evaluate the magnitude of the problem, where prioritization of resources should be made, and monitor progress toward reducing the problem. WHO and the HIS Task Force have aggressively implemented the ProMIS geocodes in their field reporting. GIS based presentations on the Polio eradication campaign have impressed both partner agencies and external bodies. WHO GIS users have now crossed the threshold from simply referencing their data into using GIS analysis to investigate their data for answers to program questions.

3.2.3. Conclusion

Every organization makes program decisions and solicits funding based on their own data, assumptions and priorities (mandates) regardless of the degree to which their data holdings are correct or complete. Donors are complicit in this process through a failure to fund information collection or insist on standardized data collection and recording procedures. However, when considering baseline data and indicators for decision-making, “A decision maker’s understanding is as important to a good decision as good data... Managers would rather live with a problem they can’t solve than apply a solution they don’t understand.”² Demographic data is central to credible analysis. While hotly disputed, such energy may be able to be focused on revising figures for selected areas. Recording all agency and NGO activities should also be a

² Joseph K. Berry, *Spatial Reasoning for Effective GIS*, (Fort Collins, CO: GIS World Books, 1995), pg. 13

straightforward exercise and will provide the core inventory of baseline data. However for any of this to be realized, organizations must adopt the geocodes as a common referent. For the common data recording standard to be achieved ProMIS must begin updating the base map. Failure to implement the geocodes and update process will render any data collection effort futile and unmanageable.

3.3. IDENTIFY MECHANISMS FOR COLLECTING AND COMPILING BASELINE DATA

3.3.1. Findings:

- There is widespread bickering amongst agencies and NGOs on data quality.
- There is significant redundancy in information collection efforts.
- Sector lead agencies have serious limitations in their ability to handle information.
- Countrywide information exists yet is not being compiled, instead new surveys are commissioned.
- Lack of UNCO or Donor (APB) support or influence promoting information management and reporting standards.

3.3.2. Discussion

Information collection in relief and development programs has multiple audiences. Relief and development communities have four key purposes for information collection:

- **Advocacy:** Advocacy in these terms is aimed at raising the awareness of potential or current donors of the seriousness and continuing needs of target populations as related to the type of programs that the advocating organization undertakes. Information supporting advocacy goals is usually generalized and descriptive in nature and while providing large indirect benefits, holds little direct value for the program manager.
- **Project Management:** Ideally such information should support the answers to questions such as how much resources, to where, how frequently, to whom, what impact, and where do I stand in regard to implementation status.
- **Policy Formulation:** Knowing what is where helps us to understand what should be achieved. Policies are usually associated with standards that quantify project implementation conditions and goals. Policy analysis allows us to interact with information collected and evaluate it against a common rubric.
- **Strategy Formulation:** Strategies communicate a common operational picture and provide guidance on the relative allocation of resources. Information supporting strategies should be derived from the analysis of policies and standards. Hence strategies can and should change in response to changes in the relative priorities assigned to the individual supporting elements.

All of these efforts rely on effective information sharing. Most often these information requirements are addressed separately and independently. Such an effort quickly overloads the limited capacity and willingness to cooperate of information collecting organizations. The objective should be to meet the varied needs from a common data

set, thereby reducing redundancy and maintaining the good will of partner organizations. However, the lack of information sharing is the most glaring shortcoming identified in After Action reports and Lessons Learned studies of all major relief operations to date. If this is the case then why can't the relief and development community seem to make headway in addressing these needs?

Other factors also condition an organizations willingness to share information. Among these are:

- Information is a non-refundable investment
- Competition for limited funding
- Territoriality
- Information Quality
- Processing capacity

A common factor in the relief and development community is that donors do not pay for assessments. Therefore the information that an organization collects is its competitive edge over other organizations in the competition for scarce funding. Given the time and resources that an organization invests to assemble a project proposal, information sharing is not always in its best interests. A related aspect is the phenomena of territoriality.

Organizations tend to mark out geographic space in which they try to exert their primacy for program implementation generally or sector specific. In the context of non-refundable investment in information collection, territoriality allows the organization to reach economies of scale for securing future funding due to their already established familiarity with the needs of that area. Territoriality also serves other aspects of program implementation. Relief organizations have varying capacities. Localizing their efforts maximizes limited capacity and reduces overhead costs of extending management over widely spaced discontinuous areas. Organizations are also sensitive to the local perceptions of their activities. Once an organization has begun service delivery in an area it finds it difficult to abruptly stop and move to another area. In this case the organization's presence and services have raised the expectations of the beneficiary population. Terminating services can result in negative reactions towards the former servicing organization resulting from unfulfilled expectations.

Information quality issues are the source of the frequent complaints that organization A is inflating its figures to attract a larger portion of limited resources. Similar claims are made concerning collection methodologies and the unreliability of the results. One analysis of such claims would find that such claims are aimed at discrediting organization A's information and hence it's ability to attract funding. Alternatively, such criticisms could be aimed at alerting potential donors that there are organizations better placed to maximize the impact of their funding. True or not, such claims affect the level of trust between organizations. For organizations to be able to share information there must be a level of trust that the information provided will not be used against them. There must also be a shared understanding and appreciation that the products resulting from information sharing and indicator analysis will directly benefit the contributing organizations. Additionally, organizations trust that they will be given due credit for information that they do provide as opposed to the collecting organization misrepresenting it as its own.

Regarding methodology issues, questions raised are usually focused on sampling techniques. Such surveys are focused on generalizing information about a given geographic area. However, most statistical analyses do not treat geographic distribution as a factor. Many field workers know that there are always pockets of higher and lower values in an area despite the results of generalized assessments. Such pockets are of particular interest to organizations that are focused on point-of-service delivery and targeting of vulnerable groups. GIS provides some answers that can simplify these issues. First GIS analysis can show that samples taken do not adequately cover the geographic space of the area under survey, i.e. mostly along main roads where accessibility is easy or clustering in specific areas and ignoring other areas. Second, GIS can calculate generalized values as in the statistical analysis yet retain geographic specificity or spatial autocorrelation. Such surface analyses would allow the visualization of patterns in data over geographic space. In this way, for example, nutritional surveys that derive Z scores, could be analyzed to show trends and high/low values much in the same manner as temperature maps show contours of value changes. Such analyses would permit enhanced targeting and assessment of communities falling under the interpolated “risk contours”, as well as permitting evaluation against other variables as potential contributing factors.

An organization’s capacity to handle information is less an aspect of willingness than of ability to share information. The most common constraint is the lack of ability to take information from collection forms or coding sheets and turn it into tabular data. Lack of computers requires hand compilation or the use of typewriters. If computer entry is available, problems occur when insufficient personnel, who are familiar with the software, are available to enter the data in a timely manner. An organizations inability to address this capacity shortfall can lead to claims that the organization is withholding or otherwise failing to meet its data sharing obligations.

3.3.3. Conclusion

In information sharing arrangements certain agreements need to be reached. Organizations will be more likely to share information if doing so does not interfere with their existing work. Similarly, organizations are more willing to share information if they see a benefit in doing so. Transfer of information will be encouraged if standard formats are adopted. Information sharing arrangements are successful if the management and participating agencies share a common vision or operational picture as to what will be accomplished. Conversely, information sharing will not succeed if the arrangements are overly bureaucratic, not responsive to partner requirements, restrict other partners from access to information, or there is a lack of trust among participating organizations. Data collection also needs to be simplified. Sector organizations should agree on a minimum set of basic data. Similarly scientific sampling surveys should utilize geocoding of collection points and apply advanced spatial analysis. These steps are not complicated nor resource demanding and the tools are already present in ProMIS. The primary issue is that agencies/NGOs implement the recording of data against geocodes to provide a standard for common reference and comparison.

4. RECOMMENDATIONS

Knowing how to use data is equally important as having good data. The last two decades have seen a proliferation of information technology and information management solutions for deriving meaning or knowledge from large information holdings. Long standing tools including spreadsheets, databases and statistical analysis computer programs have been joined by geographic information systems (GIS) for the analysis and visualization of complex information. These tools have gained significant importance in enabling managers to make more informed and timelier decisions.

4.1. ASPECTS OF INFORMATION MANAGEMENT

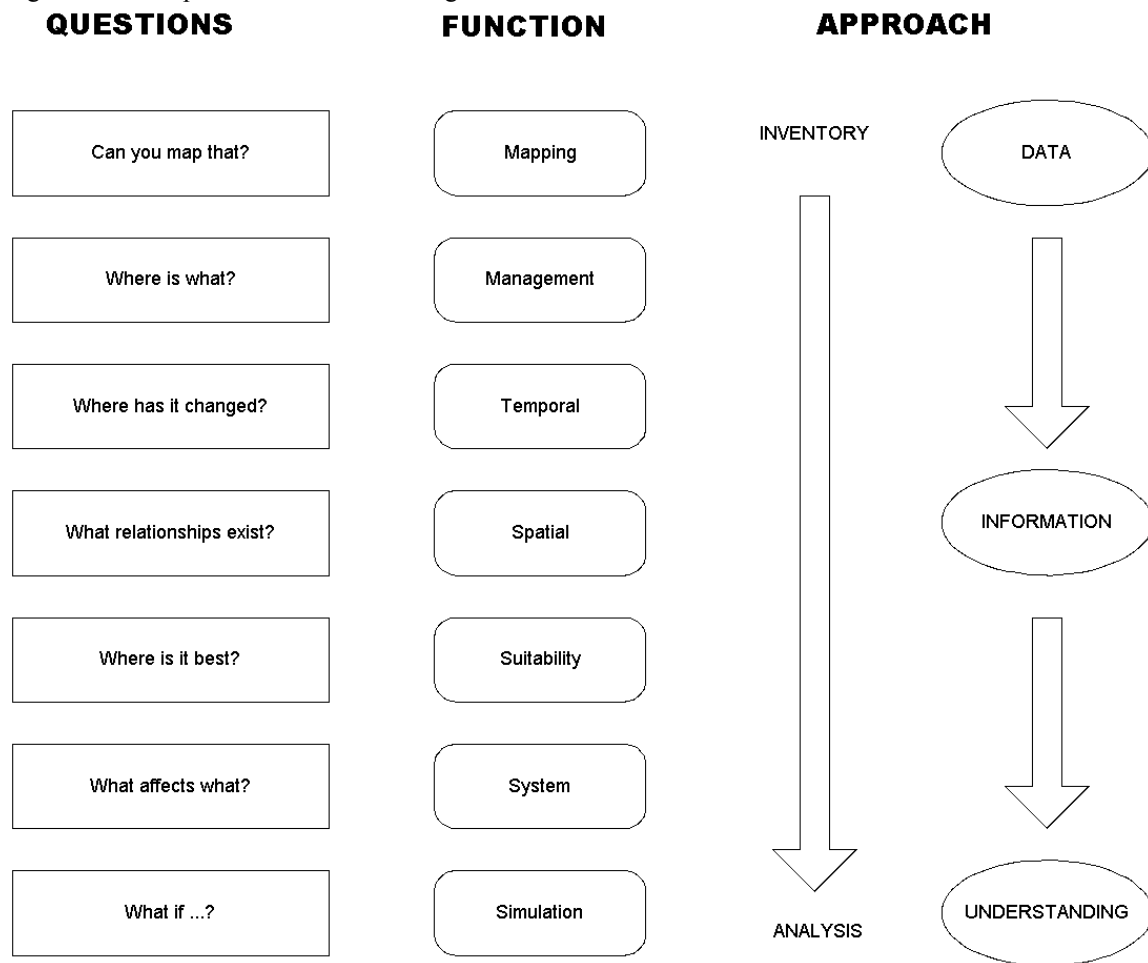
One of the major hurdles of information technology is the change that is required in human, social and organizational processes. Many are the instances of database creation resulting in little appreciable benefit to decision makers. Information systems do not tell the manager what question(s) to ask. In the past, asking serious questions were prohibited by the labor demands and shortage of skills required to analyze thousands of records. In relief and development programs this is manifested in the proliferation of large-scale annual survey generalizations and narrative description based reports “interpreting” generalized tabular data and statistics.

Information management is supposed to help us do our job better. Much as statistical analysis and pivot tables have helped people analyze spreadsheet and database information holdings in the past, GIS tools permit not only analysis but also visualization of information in format that is more closely related to reality – a map. Many a reader glosses over lengthy tables of data but is instinctively drawn to pictures or images. Pie charts, histograms and trend lines are some of the more obvious examples of visualization tools. However, these tools are not sensitive to geographic location. Moreover, most advanced statistical descriptive and predictive analyses (regressions, standard deviation, cluster analysis etc.) are not sensitive to geographic factors such as proximity. With traditional charts axis labels are usually organized alphabetically or chronologically for time series data. In a GIS presentation the data or values are represented according to their geographic location. Such a representation allows the viewer to raise intuitive questions such as: why is there a clustering of high values in these locations; are these difference due to the areas being close to cross border influences; why don’t we have data for these areas? For relief and development planners these questions and others are essential for understanding what they need to do, what impact their programs are having and what progress they are achieving.

4.2. FROM DATA TO INDICATORS

The functions referred to in figure 1 describe a progressive change in how we think about data. The Mapping and Management functions are most frequently associated with day-to-day requirements. The predominant focus is on restating inventory type information such as where are the schools in district A. The Mapping function is simply automated cartography. The bulk of customer data-based GIS products generated by ProMIS fall into this category.

Figure 1: Concepts of Information Management and GIS³



Management functions are inventory approaches that exploit the link between the digital mapping and a database of information on what is where. For ProMIS this can be fulfilled through the compilation of a georeferenced organizational activity location (GOAL) or “who’s doing what where” (WDWW) database. To be useful this should be at the level of settlement specificity. Similarly, Sector lead agencies inventorying all schools or clinics, whether or not they are supported by an agency/NGO, using geocodes can also map locations, develop catchment areas and calculate service ratios. The key constraint on assembling such a database is agreement on the minimum number of information elements, the structure/format of reporting and the standards against which data will be evaluated. Fortunately, ProMIS already has a standard record locator process called geocodes.

Temporal analysis, where has it changed type questions, marks the transition from reference and inventory-related data searches to packaging information for generating plans and policies. At this level the requirement for standards and basic data elements becomes focused and the information process moves beyond simply automating record retrieval to providing new tools.

³ Adapted from Joseph K. Berry, *Spatial Reasoning for Effective GIS*, (Fort Collins, CO: GIS World Books, 1995), pg. 43

“What relationships exist” type questions brings us into Spatial analysis and reasoning and take full advantage of GIS tools. Questions such as how does access to safe water in this location compare to that location? Are there sufficient schools to meet service area population needs? Is there a correlation between areas reporting moderate and severe dehydration and areas where mines block access to safe water? Do the risk contours generated from nutritional sampling surveys indicate settlements that may be at equal risk and therefore require selective targeting? These questions involve derived values in spatial relationships. However, uncovering these questions is difficult if there are no standards against which programs are evaluated. If you don’t know what a GIS can do differently, then you will not likely ask it to do anything differently.

Suitability and Systems questions are the core of models for planning and analysis. Where should I set my priorities? How can I maximize the impact of limited funding? Asking these types of questions allows the planner and policy maker to manipulate derived values yet still maintain reference to the geographic location of individual records in the base data. Simulation functions are just that, “what if?” questions. They involve a repetitive processing of Suitability and Systems outputs to reach a desired level of satisfaction regarding a particular decision.

4.3. RECOMMENDATIONS

So what does all this mean? It means that information management systems (databases, spreadsheets, GIS) available in ProMIS and in most agencies permit a direct link between the agency working in the village and the policy makers laying out strategies for the overall Afghan program. The central requirements are for a common referent system (geocodes), basic elements of information and sector specific standards for generating indicators. What it will cost is the time and will of organizations to implement the data structure in a disciplined manner.

4.3.1. Recommendation 1

As a matter of urgency, UNCO/APB should determine which data set will be used – the 29 or 32 province geocode sets. This determines the geocodes structure. The choice will determine how the entire structure works. Failure to decide will render information compilation or comparison unworkable. Deciding now will also simplify the implementation process. Agency and NGO database managers have noted the difficulty of applying the standard to their existing databases. They would prefer to do it once and simply append new additions to the structure, as generated through the update process.

4.3.2. Recommendation 2

ProMIS should work with RCOs, Sector agencies, ACBAR and NGOs in compiling a standard, settlement level, georeferenced organizational activity location (who’s doing what where) database. ProMIS should, in consultation, design a standard reporting format (paper and electronic). The formats of ACBAR⁴

⁴ ACBAR’s report for the most part uses the settlement name. However, given the complexities of multiple unrecorded names for each settlement and the potential for variation in spelling and transliteration, standard geocodes are by far the preferred method. Additionally, as an annual product it is of limited use. To turn it into an operational coordination tool it must be updated frequently and report at least monthly.

and the sample database field structure that is attached in Annex A should be reviewed. This database will become the main contributor of baseline information, when combined with demographic information, for analysis of the impact and progress of programs. The use of the geocodes structure should be standardized and made obligatory. Individual projects should be disaggregated down to the settlement level. For example, the common practice is to report – well drilling/27 villages. Geocodes for each settlement should be provided where the project will be/is being implemented and the number of iterations in said location. Where this departs from current ACBAR practice is that it is available for reference and update on a rolling basis, rather than annually. In-progress outputs from the database holdings should be generated monthly and distributed to all UN agencies, NGO's and donors for feedback or corrections, updates and useful modifications. The database itself can be copied and distributed to RCO's for ready reference in electronic format. GIS files can also be extracted for plotting activity location information as well as aggregating to district and Province levels.

4.3.3. Recommendation 3

Sector agencies and RCOs should collaborate with their partners in compiling countrywide datasets in accordance with the minimum requirements identified in Annex B of this report. There is a large difference between activity location databases and sector specific databases. Sector databases should record all instances of their focus, not just what agencies are doing. The purpose of this process is to be able to place activities in context of the whole problem. Doing so helps planners evaluate the magnitude of the problem, assess where needs are greater, and measure progress and impact. For example:

- All schools, not just those serviced by agencies.
- All safe water sources in a village, not just those serviced by an organization.
- All future joint surveys should require justification of survey elements and sufficient detail regarding how that information will be analyzed and the decisions that will be supported.

In selecting what to collect the users should keep in mind Occams Razor. The principle is attributed to the medieval philosopher William of Occam. The principle states that one should not increase, beyond what is necessary, the number of entities required to explain anything. In this regard organizations must agree on a basic set of common data and analyses to produce indicators that are directly related to answering the questions of what, where, how much, how effective, and what progress.

4.3.4. Recommendation 4

ProMIS should prioritize efforts on the maintenance and updating of base map layers (Province, district, village) and geocodes and facilitate dissemination to field users. This the most critical function that ProMIS can undertake. The hierarchical structure of units from province to district to settlement is the basic structure on which all data should be appended. The process includes the plotting or deletion of a location and the assignment of a sequential geocode. Comments from the field have also called attention to the lack of response from ProMIS as regards timely updates on field provided corrections. Part of this problem is a lack of dedicated staff to update this data. It is time consuming and requires sequential steps to regenerate all other derived products. Another part of the problem is the means by which the changes have been reported. Placing a pencil mark on a map and writing down the

name of the place is insufficient. ProMIS should develop a standard form for reporting changes to the base map. Once the topographic maps are made available to the field, annotations on these products will significantly speed up the process.

4.3.5. Recommendation 5

ProMIS should work with sector lead agencies, NGOs and RCO's and convene a series of sector specific workshops to define minimum essential information and analyses to support resource planning and allocation. While annex B contains recommendations for indicators, these should be reviewed by appropriate organizations. The workshops should eschew “wish-list” approaches to data and indicators. Instead the efforts should be very focused on the minimum essential data that can satisfy each technical program’s decision and resource allocation requirements and the standards and criteria for evaluating success. The aim is not to micro-manage projects, but to place the outputs on a common outlook. The effort should also seek to facilitate standardized data formats for key indicators based on analytic needs (string, number, T/F)

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ANNEX A – WHO’S DOING WHAT WHERE DATABASE

The following paper was prepared for the Humanitarian Community Information Center established in Kosovo by OCHA. It is attached to provide a point of departure for recoding agency activities.

DISCUSSION PAPER

[N.B. This paper was prepared, following a meeting with the IACU, where they expressed interest in developing a tool to better capture the extent and nature of humanitarian operations in Kosovo.]

Humanitarian Database: Who’s Doing What Where

1. Requirements:
 - Coordination tool relevant to implementing organizations, policy and sectoral coordinators and Donor strategy/ resource allocation process.
 - Simplified format to encourage submission of information
 - Focus on core information that most organizations will be willing to provide
 - Not an internal project management information system
2. Suggested Fields
 - a. Main Project Fields
 - **Project #:** automatically generated. This a reference number for locating records within the database
 - **Implementing Organization:** name in short form (should be keyed to a separate short/long name table
 - **Donor(s):** major donor funding source (also keyed to the long/short name list)
 - **Description:** description of project objectives such as: “rehabilitate 15 classrooms and 3 administrative rooms in the Vlastica school” or “distribute warm room/dry room kits”.
 - **Sector:** general sector of intervention. See sector discussion below.
 - **Sub sector:** specific task of intervention. See sector discussion below.
 - b. **Project Activity:** Should be a sub-table to allow multiple location activity entries. Most organizations implement projects in multiple locations. This sub-table is intended to capture this information under a single project heading rather than multiple entries. See attached spreadsheet example. Suggested fields:
 - **Location:** P-CODE entry is ideal, but don’t reject place names. Possible automatic generation of Opstina, locality and cadastral, KFOR AOR and UNHCR FO-AOR correlations

- **Implementing Partners:** short or long name
 - **Activity:** specific description of activity at this location, e.g., renovate clinic, repair 4 wells, renovate school.
 - **Beneficiaries:** estimated direct/indirect
 - **Status:** **Assessed**-organization has completed an assessment, **planned**-organization has submitted a proposal for funding to a donor; **Pending:** funding has been approved and project will commence as planned; **Ongoing**-project has started, **Suspended**-project has been suspended pending review; **Completed**-project has been completed (could be a check box field)
 - **Start Date:** date project at this location started
 - **Suspended date:** date project at this location suspended
 - **Completion Date:** date project at this location completed
 - **Comment:** additional notes or references to other databases
3. Suggested Sectors/sub-sector
- **HEALTH:** infrastructure(types of entries includes hospital, health house, Ambulantas and Mobile), immunization (coverage rates by municipality, age-group, antigen. Some of this data might be better covered by WHO or included in the comment field), primary health care delivery, tertiary health care services, reproductive health, trauma counseling, nutrition, surveillance (mortality/morbidity rates by location in the comment or linked to separate WHO surveillance table)
 - **FOOD AID:** general distribution, vulnerable group feeding, institutional support, FFW. Include fields or add to comment field for tracking ration=kcal/person/day.
 - **SHELTER:** rural winterization, urban shelter, host family, collective shelter
 - **WATER:** municipal purification, rural well repair. Include in comment field data on water supply by point/system of liters/person/day.
 - **SANITATION:** garbage removal, public health education, latrines (include type family/communal)
 - **DEMINING:** mine/uxo clearing, mine sweeping/demarcation, mine injury reporting, mine/uxo awareness information
 - **GOVERNANCE:** support to civil authority, technical assistance, electoral support, civic education, Human Rights monitoring.

- **INFRASTRUCTURE:** road repair, bridge repair, supply of institutional heating fuel, repair of utilities
- **EDUCATION:** school renovation, teaching materials, teacher stipends
- **CIVIL SOCIETY:** media support, capacity building, public information
- **AGRICULTURE:** Wheat seed distribution, vegetable seed distribution, seed multiplication. Fertilizer distribution, pest/disease control, equipment repair, surveillance, technical assistance, coordination/information

4. Other suggested Linked Databases

- Location: this is already in place with the p-code system
- Population: Running table of population estimates over time. Could also be broken down by ethnicity, age-group and gender. The last two could be extracted from various organizational reporting i.e. food aid registration, school registration, immunization programs, human rights reporting.
- Organizational long-name/short-name
- Personnel Locator: organization, name, position, duty location, contact information (office/residence address, telephone, cell phone, fax). Could include passport number for security/evacuation purposes.

Sector Specialists have indicated a requirement for greater detail in surveillance data. Through the local GIST we should be able to set up monitoring responsibilities with periodic updates extracted from these data sets.

Suggested Field Office Coordination Report Format
FO AOR sample report structure

Municipality

Location

Sector

Sub Sector

Proj#	Imp Org	Imp Prt	Proj Activity	status	comment
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Suggested Sector Coordination Report Format
Sector

sub-sector

Municipality

proj#	Donor	Imp org	Imp Prtn	Project description	sort pcode	Status
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Proj#	Imp Org	Donor	Desc	Sector	Subsector	Location	Imp Prtnr	Activity	Beneficiaries	Status	Start Date	Susp Date	Compl Date
1173	IRC	USAID	repair rural wells	Water	Rural water	01030	MTS	repair 5 wells	1500	Ongoing	990815		
						01031	MTS	rpr 3 wells	900	Pending			
						01032	MTS	rpr 6 wells	1800	suspended	990815	990816	
1178	Elektrokosova	USAID	provide 14 transformers(12x100kVA, 2x160kVA to elektroskova-Gjilane, Vitina & Kamenica Municipalities	Infrastructure	Electric Utility								
						04024		install transformer		Ongoing	990811		
						04082		install transformer		Ongoing	990811		
						04001		install transformer		Ongoing	990811		
						04071		install transformer		Ongoing	990811		
						04061		install transformer		Ongoing	990811		
						04052		install transformer		Ongoing	990811		
						04057		install transformer		Ongoing	990811		
						10053		install transformer		Ongoing	990811		
						10125		install transformer		Ongoing	990811		
						10135		install transformer		Ongoing	990811		
						10113		install transformer		Ongoing	990811		
						26045		install transformer		Ongoing	990811		
						26007		install transformer		Ongoing	990811		
						26002		install transformer		Ongoing	990811		

Sample WDW data sheet

ANNEX B – MINIMUM DATA REQUIREMENTS AND SUGGESTED INDICATORS

BASIC CONCEPTS

Unless otherwise stated all basic data includes geocoding down to the settlement level. The following are suggestions only. They are drawn from observed practices and or analyses as identified in practice or in documentation from agencies in Afghanistan. Others are drawn from international standards such as the SPHERE Project. As interest and application builds agencies can conduct sector Information Needs Assessments (INA). The INA is a detailed process that starts with identifying the types of question for which answers are needed. Then sketch map layouts are drawn showing what and how information should be displayed and the types of analysis of the data required to generate each layers values. Then the information requirements and formats are identified which will be required to support these analyses. Finally map layers are tabulated against component data sets and aggregated to determine the frequency of demand for each type of base data. Such a study was beyond the time constraints of this consultancy, yet should form a part of the ProMIS facilitated support to sector and policy planners.

Using Indicators: Viewing indicators is different from using indicators. Once an indicator has been calculated, viewing the results provides a view of relative needs. To transform this into policy and program inputs the manager must ask how this condition can be improved. Solutions can be direct or indirect in nature. Decreasing malaria could involve distribution of treated bed nets, reducing standing water, drainage issues etc. In inventory issues such as schools, clinics, and safe water a policy goal should be set. For example access to safe water should be increased by 10% overall this year. The actual requirements can then be simply calculated based on the underlying data that produced the original access ratio. The degree to which the original information recorded was geocoded determines the degree of specificity of the results.

Data standards: Aside from survey sampling methodology the single biggest constraint on compiling shared data is format. Most databases designed for relief and development are primarily used for reference purposes, not analysis. Hence data fields are compact and make free use of multiple text entries in each field. Such a structure is prohibits effective and timely analysis. Moreover, if different organizations use different field headings, types and entry values, the transaction costs are relatively high, discouraging effective information sharing. Some basic tips in designing databases include:

- Use a common referent. In the case of ProMIS this means the geocode. Additions to the geocode such as facility number (clinic, schools) should be determined collectively and in cooperation with the ProMIS.
- Better design features would break multiple-choice answers into separate true/false or binary variable fields.
- The binary variables are easily exported to statistical analysis programs as well as lending themselves well to applied weighting factors.
- Avoid text entries for everything except where they cant be avoided (place names etc.)
- Numerical units should also be standardized. While Afghanistan has a degree of local variation in estimating the size of a *Jerib* (land area unit) and *seer* (volume unit). Reporting organizations should use the local profile to convert the measures

into metric measurements. Numerical fields should be standardized in terms of numbers of decimal places and units of measurement.

- Document the structure with details on each field regarding contents, units and meaning. If the field is a calculation, define it.
- Don't format databases or spreadsheets for paper publication with borders and shading. For compilation purposes all such features must be stripped off before joining to other databases.

While the implementation of a GIS can require extensive preparation of existing data, once completed the addition of new data is rather simple and straightforward. "A GIS can solve problems and pose new questions, but it can also promote illusion. Lay users can easily be seduced into believing two things that are not always true:

- That all of the maps produced with data from the GIS will be of equal accuracy, and
- That data that is associated geographically has a cause and effect relationship in the real world."⁵

Hence deciding on what the significant classes are for rendering the data is as or more important than the data alone.

Defining Units: Ideally, all data should be recorded at the level of the settlement. Given that most organizations implement point-of-service projects, this should not be a problem, excluding areas where there are no agencies/NGOs working. Such a procedure is an inventory approach to data management and a necessary first step. Inventory data management is insufficient to address questions of sufficiency or relative need and prioritization. To answer these questions we need to be able to analyze units and visualize geographic units according to indicator values. For most analyses, aggregating values at the district level is sufficient. However, to look at micro level comparisons between communities and or service areas that are smaller than a district requires another kind of tool.

Barring the availability of a cadastral survey that allocates land ownership and community level geographic association, there are artificial tools that can be applied. The concept of Thiessen Polygons is a well-developed GIS tool for defining geographic space among points. A GIS can create Thiessen polygons simply and efficiently from existing settlement data. The basic concept is that each point is evaluated in relationship to its nearest neighbors. A polygon is then defined with boundaries exactly fifty percent of the distance between each point and its neighbors. The resulting mosaic of polygons may not bear any relationship to reality on the ground except that each polygon is centered on a settlement point, but it serves as an effective approximation and visualization tool. Thiessen polygons can be aggregated into large units through simple merge functions. For example, a C3 clinic that has recorded where its outpatients are coming from could use merged Thiessen polygons to define its service area. Such a tool helps define what is covered and what is not covered. In this way, questions of service sufficiency and approaches to solving the problem (build new clinics or recruit staff to extend outreach) can be addressed in a logical and transparent manner. Similar approaches can be used for defining school districts, Shura administered areas, canal/karez maintenance districts and many others.

⁵ John O'Looney, *Beyond Maps: GIS and Decision Making in Local Governments* (Redlands, CA: Environmental Systems Research Institute, Inc, 2000), p. 31.

1. DEMOGRAPHICS

Primary ranges of demographic data collection by major survey standard

Ranges/ Survey Tool	<1	1-4	5	6-9	10-14	15-19	20-44	45-59	>60	Preg	Lact	Fem HH	M/F
Hasty Health													
Health Surveillance										X	X	X	
Demographic Profile										X	X	X	X
Nutrition										X	X	X	
Education													X

The table above illustrates why, for the most part, agencies have been unable to cooperate in collecting demographic data. The shaded portions indicate age ranges of interest for the survey tools. The shadings were chosen for contrast purpose and carry no other significance. The age breakdowns and additional classes represent ranges that permit a minimum level of comparison. The major area of contention is the range for ages 6-19 that is the primary focus for education services yet of marginal use for health and nutrition organizations.

In estimating populations a demographic profile can be used to extrapolate from a known subset of the population. The standard profile (Sphere) is:

- Age 0-4/12.37%,
- 5-9/11.69%
- 10-14/10.53%
- 15-19/9.54%
- 20-59/48.63%
- 60+/7.24%
- pregnant/2.4%
- lactating/2.6%
- male/female 50.84%/49.16%.

Such profiles do not always fit populations that have come through prolonged crises. Hence interagency efforts should focus on common elements of collection and of demographic profiling as part of a shared information collection effort. Particularly in areas where organizations have long experience, sampling at the settlement level should be part of an ongoing process. Current population estimates have focused on average household size. Demographic profiling and selective head counts can balance the sometimes inflationary action of average household size. Moreover, Given the widespread use of Household Surveys, such profiles should be incorporated in all agency surveys.

Demographic data is essential for evaluating the impact service delivery programs. To make analyses comparable a standard population estimate should be agreed upon. Equally important is the agreement on major population units (county, province, district) as well as a mechanism for selectively revising the figures based on periodic updates from field organizations. The current practice of extrapolating from earlier census data is not useful in capturing displaced population. It is useful for long term planning of services for populations expected to return to an area. Such information could help decide whether funds should be expended in rebuilding a clinic versus providing temporary/mobile clinic services.

2. EDUCATION

Education systems are generally structured on an area of service principle. Hence in gathering data it is always important to determine the service area for schools. An obvious way of collecting this data is to document where (settlement code) the enrolled students are living. Such a process will help focus on progression issues from primary to secondary levels.

Key aspects of support to the education system are the issues of quantity and quality. Quantity means answering the questions of is there access and sufficient resources. Quality refers to issues of curriculum, teacher training, and learning environment. Most education programs deal with: build/renovate, text book delivery, teacher training, and expanding opportunities for education for boys and girls All data should recorded at the individual school name/settlement level to facilitate varying levels of aggregation (i.e. service area, district, province). Data for formal and informal schools should be recorded because the idea is to capture the degree of access to education. Suggested minimum elements of information deal with varying degrees of detail on students, teachers, classroom/infrastructure and teaching materials.

Education Indicators	Data requirement
Enrollment Ratio by grades (1-3, 4-6, 7-12)	Students enrolled by grade
	Total school aged population by grade
Female Enrollment ration by grades (1-3, 4-6, 7-12)	Female students enrolled by grade
	Total female school aged population by grade
Male enrollment ratio by grade (1-3, 4-6, 7-12)	Male students enrolled by grade
	Total male school aged population by grade
Student/teacher Ratio (grades 1-3)	Teachers per grade per school (grades 1-3)
	Students per grade by school (grades 1-3)
	Ideal student/teacher ratio
Student/teacher ratio by subject (grades 4-6)	Teachers per Subject by school (4-6)
	Students per grade by school (grades 4-6)
	Ideal student/teacher ratio by subject
Student/teacher ratio by subject (grades 7-12)	Teachers per Subject by school (7-12)
	Students per grade by school (grades 7-12)
	Ideal student/teacher ratio by subject
Percent qualified teachers	Number of teachers at each level of qualification
	Ideal percentage of qualifications
(Grade 1-3) textbooks per student ratio	Number of text books per grade (1-3)
	Number of students enrolled (grade 1-3)
	Ideal student/textbook ratio by grade
(Grades 4-6) textbooks per student per subject ratio	Number of text books per grade per subject (4-6)
	Number of students enrolled by grade (grade 4-6)
	Ideal student/textbook ratio by subject
(Grades 7-12) textbooks per student per subject ratio	Number of text books per grade per subject (7-12)
	Number of students enrolled by grade (grade 7-12)
	Ideal student/textbook ratio by subject
Teacher /Classroom ratio	Number of Classrooms per school
	Number of teachers per school
	Ideal ratio
Student/classroom ratio	Number of Classrooms per school
	Number of students per school

	Ideal ratio
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Indicators are used to evaluate the degree above or below the standard or ideal ratio. Policy goals evaluate shortfalls and set targets of percent increase for the program year or other period. Use this value to revisit base data to determine per item counts required by the target.

3. FOOD AID

WFP's VAM unit uses a documented Emergency Food Needs Assessment process. The Assessment uses a rather broad-brush approach and could make greater use of the GIS software that they own. In particular, the survey generalizes too much over too large of an area. Use of geocoded sample points can help increase geographic coverage and permit greater ease of identifying trends and patterns of geographic food security issues. Such an approach can be more easily integrated with other sector factor analysis. Recommendations include:

- The Household interview portions should incorporate a demographic profile component to the existing factors of female-headed households and average family size.
- WFP should evaluate the benefits of expanding the data collection process and seeking greater geographic specificity. Perhaps encouraging NGOs to carry out surveys to standard as a way of increasing coverage.
- WFP should implement the common geocodes for recording data
- WFP should standardize on an agreed system-wide population estimate.
- VAM should approach ProMIS to coordinate GIS training

4. HEALTH

There are two aspects of health related data: epidemiological monitoring and health care delivery

WHO and the Health Information System Task Force have just launched a surveillance system for collecting outpatient epidemiology data. The system is well structured and uses well established methodology for evaluating data. Using geocodes to record the locations of data collected when combined with GIS analysis will allow the evaluation of morbidity or prevalence rates in geographic context as well as against other factors similarly aggregated. Such an approach will contribute to a more comprehensive public health management program

Health care delivery has yet to be evaluated. A 1994 study "Project Framework: Minimum Primary Health Care for Afghanistan" detailed a hierarchical health care delivery system providing expected service capacities and staffing requirements. Once an inventory is completed of health facilities, service ratios can be calculated. To be most effective catchment areas should be estimated by reviewing HIS surveillance forms for village of residence of those attending the facility. Before catchment areas are defined, district service ratios can be calculated using the same approach, but without the individual facility specificity.

The identified standards were

- C-3 Clinic/Basic Health Post – serves 5,000 population
- C-2 Clinic/Basic Health Centers – serves 30,000 population
- C-1 Clinic/Comprehensive Health Centers – serves 50,000 population
- H-3 Hospital/ Primary Health Care Hospital – serves 150,000 population

Health Care Delivery Indicators	Information Requirements
District Primary Health Care Service Ratio	Population of area
	Number and type of health facilities in area
	Optimal capacity standard

Basic formula:

Optimal capacity standard/(Population/number of facilities) = service ratio
 Values over 1.0 have excess capacity, values under 1.0 have a capacity shortfall. This procedure can be used for district, province as well as catchment area.

This same formula can be applied to other standards frequently used such as TBA's per capita or Physicians per capita. The greater the geographic specificity of the data, the greater the precision in analysis

5. WATER

As with other point-of-service issues the water sector should compile all safe water points in a settlement. Compiling the DACAAR and SCA data sets should provide records for over 20,000 wells and is a good starting point for further data collection and survey. The current UNICEF survey appears to be ignoring these large extant data sets in favor of a new survey.

The primary issue is access to safe water. The SPHERE project provides a standard of 250 persons per safe water point, at a maximum distance of 500m, with a flow rate of 0.125 L/sec. By inventorying all safe water points in a community relative access can be calculated against the standard. Surveys that collect a household interview should also include a basic demographic profile and head count by age/gender groupings, as noted above. For example, the UNICEF water survey is estimating family size, counting children under 5 and Female headed households. Therefore, adding a few more categories to demographic collection would significantly contribute to the overall understanding of population dynamics in Afghanistan.

Safe Water Access Indicators	Information Requirements
Safe water point access ratio	Population of area
	Total number of safe water points in the area
	Optimal capacity standard (250person ea)
Safe water access (piped water)	Population of serviced area
	Minimum water requirement per person
	System flow rate
Sanitation Access Indicators	
Access to excreta disposal	Population of area & number of households
	Number of standard toilets available
	Refined by number of households without a toilet

If the skill is present to monitor water quality issues, these indicators can be built on to include technical measurements of water depth, flow rate, contaminants, taste and conductivity. The SPHERE project includes minimum internationally accepted values for the requirements.

Sanitation

The basic elements of sanitation beyond education initiatives include excreta and solid waste disposal. Most surveys to date have paid little attention to these issues. One survey even discounted the importance of proper sanitation, justifying defecation fields as sufficient. The Sphere Project provides the following standards for:

- Excreta disposal
 - 20 persons per toilet aggregate and by household or segregated by sex,
 - Within 50m or 1minute walk,
 - 30m from water source
 - 1.5m above water table
- Solid Waste disposal
 - No dwelling is more than 15m from a refuse container or household refuse pit, or 100m from a communal refuse pit.
 - One 100 liter refuse container is available per 10 families where domestic refuse is not buried on site.

Such factors address the basic elements of information regarding where and what are the needs. Quality issues can be more technical in detail and should reflect a sector wide consensus of what is feasible to collect and the frequency of monitoring. With these basic elements needs may be defined without micro managing the programs of individual organizations. More exhaustive standards are available in the Sphere Handbook available at the UNOCHA Program Office. Electronic copies are also available.

6. GOVERNANCE

The existence and functioning of organized local administration such as Shura and Canal/Karez maintenance districts could serve as an indicator of strengthened local governance. Using geocodes, filed data collection can specify the seat of the body and the settlements comprising the administrative area, which can then be turned into a mapable unit of analysis. Such units could also become level of engagement for integrating relief and participatory community development.

7. AGRICULTURE

Coordination in the agricultural sector appears to be heavily fragmented. While agriculture covers crops (rain-fed and irrigated), livestock, forestry, illicit crop reduction, and erosion control issues, no single agency has complete responsibility. Few of the organizations engaged in this sector were available for consultation. Strong crosscutting issues exist, but only limited intersectoral links exist, among agriculture, water and sanitation, demining, and food security, refugees, disaster mitigation and governance. The major constraint is a lack of a picture as to who is doing what where at all. Ideally, such a database should detail down to the settlement level. Compiling and maintaining such a database should be of primary importance. Without such a database it is relatively impossible to determine the actual extent of agricultural interventions.

Between crops and livestock the important issue is improving the well being of the household. For food crops this is defined as increasing the amount of surplus production over subsistence requirements and the ability to market the produce. While some projects address farmers' cooperatives, few projects listed in the appeal appear to address food preservation or farm to market roads and transport.

For livestock a balance must be struck between increasing herd size and the carrying capacity of the grazing land (water, biomass) and other sources of fodder. Sphere provides an estimate of 20-30 liters/day for large animals and 5 liters/day for small animals. Rough estimates should be able to be calculated using 1993 (FAO) Land cover analyses and historical hydrography data. Such carrying capacity issues will necessarily vary over the geography of the country. Such spatial analysis could be effected quickly with the use of GIS tools.

Keeping the livestock healthy and reducing diseases can also do with some adjustments. Total cases are reported currently along with total vaccinated animals. Besides absolute numbers such data does not help in improving service delivery and coverage. Better measures would look at prevalence rates over sampled geographic units. Similarly FAO has been experimenting with off take rates and other measures of evaluating the state of herds. If these prove useful, serious attention should be given to expanding the use to other independently funded agencies.

A workshop regarding standards and goals for the agricultural programs should be convened. The focus should be on reaching general agreement on basic data collection and common indicator analysis that can be collected uniformly by all organizations engaged in such activities.

8. REFUGEES

Data collection on refugees is fractured along country office lines. UNHCR Afghanistan, Pakistan and Iran all have different information collection procedures and standards. UNHCR Afghanistan efforts are also split between returnee registration and returnee monitoring carried out under the Protection function. Data for each is handled separately and independent staff completes data entry. The key questions being sought by UNHCR are related to:

- Where are the returnees heading?
- What are their needs in those locations?
- What level of services already exist in those areas?
- What areas of return are relatively underserved?

Current data collection procedures lack sufficient geographic detail to facilitate targeting below the level of the Province. Given the wide variation in access to basic services across most provinces, there is a need for greater detail on exact locations of return. However, since ProMIS has been unable to provide up to date settlement level lists, such recording has not been possible. UNHCR acknowledges the fact that it has been unable to provide timely updates of returnee destinations. Some delays average almost three months. Sub-Offices responsible for collection and data entry do not have the skill or sufficient trained staff to process the information collected. Accurate and timely reports are essential for coordinating support service ranging from shelter and agriculture to mine clearance. Standards used in evaluating access to services do not seem to be those used by the respective sectors.

Data collection suggestions

- Collect complete demographic detail (profile and headcounts by age groups)
- Use ProMIS geocodes to pin point returnee destinations/current locations down to settlement level.
- Use sector standards for reporting on access to services
- Prioritize enhanced data entry capacity
- Evaluate returnee impact in terms of relative ratio to existing population

External data set requirements

Georeferenced organizational activity location database (to be developed by ProMIS with ACBAR, RCOs, Sectors)

ANNEX C - CONSTRAINTS ON REALIZATION OF THE ORIGINAL ProMIS STRATEGY

Without management support, without a knowledgeable project manager and without any organized means of encouraging sector organizations to compile their data, ProMIS has been wrongly criticized for failing to deliver on the project objectives. Such criticism is poorly placed. Data collection and maintenance are agency/NGO dependent. The original ProMIS project plan was thrown off schedule and early expectations have gone unfulfilled. While Phase One of the project entailed the delivery of equipment and software to establish the core database framework, Phase Two was to focus on information management services and support. Follow through on Phase Two was abandoned under the first local Project Manager, with certain decisions actually undermining the ability of ProMIS to meet its goals. Specifically:

- ProMIS management policy against adjusting the base-map from field observed changes (new towns, changed names etc) has stifled ProMIS's efforts to establish a common data recording and reference system. Hence, there has been limited success in promoting information sharing.
- Too much focus on servicing ProMIS software vs. promoting data collection networks (human not electronic).
- ProMIS staff are burdened with repetitive routine cartographic tasks and are unable to devote time to information management services.
- Insistence on ProMIS logos in the lead position on products has alienated would be data providers by denying them due credit.

In fulfillment of Phase two, ProMIS should prioritize and expedite the following tasks.

- **Expand Client Services and Support.** A reorientation of current work from servicing the software to servicing the clients. This requires increasing staff efficiency, providing rapid updated information and feedback to field users and aggressive marketing and education of program managers on the benefits of investing time in information management. Notably:
 - ProMIS should prioritize efforts on the maintenance and updating of base map layers (Province, district, village) and geocodes and facilitate dissemination to field users.
 - ProMIS should focus core efforts on information management service and support. The main efforts should be on:
 - Secure a program manager to effectively market the ProMIS services, ensure key decisions are addressed and serve as a focal point for information standards and marketing/promoting alternative analysis options.
 - Becoming an internal consulting body to the Afghan program, helping organizations to review data collection and analysis practices;
 - With sector lead agencies, NGOs and RCO's convene a series of sector specific workshops to define minimum essential information and analyses to support resource planning and allocation.
 - Promote/facilitate standardized data formats for key indicators based on analytic needs (string, number, T/F);
 - Promote/facilitate standardized field headings in databases for common reportable items;

- Archive of time series data sets from organizations;
 - Technical assistance to supported agencies/NGOs on database design and survey design data formats;
 - Establish GIS user groups to identify training needs and facilitate skill enhancement.
- **Distribute CD based Topographic Maps.** Scan and register 1:100,000 scale topographic maps onto CD for distribution to field organizations. This will not only provide RCOs and field based organizations with ability to print detailed cartographic products for navigation and orientation purposes, but also relieve ProMIS staff of routine cartographic product demands and facilitate more detailed feedback on changes to the base map. ProMIS holds a full set of 1:100,000 scale Topographic maps for Afghanistan. These maps can be scanned and packaged with a free GIS viewer onto a compact disc, for distribution to field organizations. Shape files for provincial, and district boundaries as well as geocoded settlements should be provided as overlays to facilitate verification and orientation. Advantages include:
 - Providing a “mapping-on-demand” capacity for field-based organizations to improve agency and NGO field operations management, monitoring and assessments.
 - Direct application for improving Mine Action Planning (they currently use a very inadequate 1:500,000 level of topography).
 - Can more easily bound settled areas to give a true picture of population densities.
 - Routine customer product orders for ProMIS would also be reduced by at least 60 percent, thereby freeing up ProMIS GIS staff to focus on rapid updates of the base map and more direct support to organizational analytic products.
 - Annotated Printouts would also significantly improve the quality and usability of field feedback to ProMIS for updating the base map.
 - A cost estimate was submitted by a Pakistani Engineering firm with an A0 Scanner and strong ties to ESRI. Cost per sheet for scanning, registering and indexing was 5,000 Rp. For 256 map sheets that comes to approximately \$23,000. During debriefing a number of organizations indicated a willingness to cost share such a project (UNICEF, UNOCHA/MAPA).
 - **Senior Management Ownership:** Information An Information Standards Committee (UNOC/APB level), as originally proposed in the ProMIS project documentation, be constituted and convened regularly to review and promote information management requirements, address data entry bottlenecks, reduce redundancy and track sector efforts of standardizing information requirements, compilation of member data sets and setting policies and standards. Representation should be drawn from UN agency, UNCO, NGO and Donor Communities. Ideally it would also serve as a both a thematic group and as a working group/sub-committee of the APB. Standards should be endorsed by APB and UNCO and made part of all future funding arrangements. Major tasks of the committee include:
 - Coordinating policy on data standards
 - Promoting electronic tabular reporting of data using approved standard fields.
 - Promoting measures for reducing redundant/duplicate data collection

- Identifying possible joint endeavors to address cross sector information standardization, i.e. standard population figures.
- Finding collective solutions to bottlenecks relating to lack of data entry capacity.
- Reviewing additions/modifications of the base map and geocodes allocations.
- Standards for graphics (symbols, line types, colors or black and white(field users don't have color printers hence the need for intelligible B&W rendering)
- Review for possible conflict/confusion on supplementary coding systems (school number/catchment area, clinic number/catchment area) or potential expansion of geocodes to account for future increase in units of analysis (Alaqadari, mantaqa, shura areas, Canel/karez maintenance districts for example).⁶

⁶ After interagency discussion, geocodes in Kosovo were expanded to include codes for polling districts (elections), police patrol areas, urban neighborhoods and rural subdivisions smaller than the district. The decision was made jointly such that the codes would be discrete and acknowledged by all organizations.